

Trade Study TS-006

Ground Control Station Human-Machine Interface (HMI) Architecture Selection

Project: Instrument Case Modernization for GCS as a subsystem of TOMTOM

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1. Objective

The objective of this trade study is to evaluate candidate Human-Machine Interface (HMI) architectures for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM) Ground Control Station (GCS) and select the configuration that best satisfies the project's operational, technical, and ergonomic requirements. The HMI architecture defines how the operator interacts with the Ground Control Station during mission planning, aircraft control, payload management, telemetry monitoring, and mission execution.

Because the Ground Control Station serves as the primary operator workstation throughout all mission phases, the selected HMI architecture must promote efficient operator workflow while minimizing workload, reducing the likelihood of input errors, and supporting rapid interpretation of mission-critical information. The architecture should also provide sufficient flexibility to accommodate future software enhancements and additional mission capabilities without requiring significant redesign of the operator interface.

In addition to supporting current prototype objectives, the selected HMI architecture should provide an intuitive and expandable platform suitable for future autonomous operations, sensor integration, and mission planning applications while maintaining compatibility with the overall Ground Control Station design philosophy.

2. Design Requirements

The Human-Machine Interface shall provide the operator with an intuitive, efficient, and reliable means of controlling the Ground Control Station during all phases of operation. The selected architecture must support mission planning, flight monitoring, telemetry visualization, payload control, and system configuration while minimizing unnecessary operator workload.

The HMI should provide sufficient flexibility for future software expansion while remaining compatible with Linux-based mission software and the physical layout of the Ground Control Station. Accessibility, maintainability, and operator efficiency were considered primary design objectives throughout the evaluation process.

Requirement	Value
Linux Compatible	Required
QGroundControl Compatible	Required
Supports Keyboard Input	Required
Supports Pointing Device	Required
Portable	Required
Ergonomic	Required
Expandable	Required

Easy to Maintain	Required
Budget	<\$300 Preferred

3. Alternatives Considered

Alternative A

Laptop Integrated Interface

The integrated laptop interface utilizes the existing laptop display, keyboard, and touchpad as the primary operator interface. This architecture minimizes additional hardware while maintaining compatibility with existing mission software.

Advantages

- Lowest acquisition cost
- Minimal hardware
- Simple integration
- Compact

Disadvantages

- Limited screen size
- Reduced operator flexibility
- Difficult simultaneous monitoring

Alternative B

External Monitor with Keyboard and Mouse

This architecture separates the display from the primary input devices by utilizing an external monitor together with a dedicated keyboard and mouse.

Advantages

- Larger display area
- Improved ergonomics
- Better multitasking
- Flexible layout
- Easy maintenance

Disadvantages

- Additional hardware
- Increased power consumption
- Higher acquisition cost

Alternative C

Touchscreen Interface

A touchscreen interface combines display and operator input into a single device.

Advantages

- Simplified interaction
- Reduced hardware
- Compact
- Modern interface

Disadvantages

- Reduced precision
- Less effective long sessions
- Higher cost
- Limited optimization

Alternative D

Hybrid Interface

The hybrid architecture combines an external monitor with a conventional keyboard, pointing device, and optional game controller.

Advantages

- Maximum flexibility
- Multiple input methods
- Expandable
- Excellent efficiency
- Adaptable

Disadvantages

- Highest complexity
- Largest footprint
- Higher cost

4. Evaluation Criteria

The candidate Human-Machine Interface architectures were evaluated using a weighted decision matrix based on operator efficiency, ergonomics, expandability, implementation complexity, lifecycle cost, and compatibility with the Ground Control Station architecture.

Operator efficiency received the highest weighting because the Human-Machine Interface directly affects mission execution, operator workload, and situational awareness. Ergonomics and expandability were also assigned significant weighting to ensure the selected architecture supports prolonged operation and future software development.

Criterion	Weight
Operator Efficiency	25%
Ergonomics	20%
Expandability	20%
Cost	15%
Maintainability	10%
Ease of Integration	10%

5. Decision Matrix

Criteria	Weight	Laptop	External Monitor	Touchscreen	Hybrid
Operator Efficiency	0.25	3	4	3	5
Ergonomics	0.20	3	5	3	5
Expandability	0.20	3	4	3	5
Cost	0.15	5	4	2	3
Maintainability	0.10	4	5	3	4
Ease of Integration	0.10	5	4	4	4

Each alternative was evaluated using a qualitative scoring scale ranging from one to five. The weighted decision matrix provides a structured and repeatable methodology for comparing candidate interface architectures while reducing subjective bias during engineering decision-making.

6. Weighted Scores

Alternative	Total Score
Laptop Integrated	3.50
External Monitor	4.35
Touchscreen	3.00
Hybrid Interface	4.55

Weighted scores were calculated by multiplying each criterion score by its weighting factor and summing the results. Engineering judgment, future software expansion,

implementation flexibility, and lifecycle maintainability were also considered in the final selection.

7. Discussion

The hybrid Human-Machine Interface achieved the highest weighted score and was selected as the preferred operator interface architecture for the TOMTOM Ground Control Station. Although this configuration requires additional hardware, it provides significantly greater operator flexibility, improved ergonomics, and superior multitasking capability.

The selected architecture allows the operator to employ a dedicated monitor, keyboard, pointing device, and optional game controller according to mission requirements while maintaining compatibility with Linux-based mission software and future system enhancements. This modular approach improves maintainability, simplifies subsystem upgrades, and provides flexibility for future mission applications.

While simpler configurations reduce hardware requirements, they also reduce operator efficiency and long-term expandability. The hybrid architecture therefore provides the best overall balance between usability, flexibility, implementation effort, and lifecycle value.

8. Decision

Selected Alternative:

Hybrid Human-Machine Interface

Rationale

- Operator efficiency
- Ergonomics
- Expandability
- Maintainability
- Ease of integration
- Long-term flexibility

The hybrid Human-Machine Interface establishes the operator interaction baseline for the TOMTOM Ground Control Station and provides an ergonomic, expandable, and mission-capable platform supporting prototype integration, subsystem verification, and future operational capability development.